

JUNWEN LIU

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APPOINTMENTS AND EDUCATION

The University of Hong Kong, Hong Kong Aug. 2026 – Present

Postdoctoral Fellow

Supervisor: Prof. Zhigang Bao

Fudan University, Shanghai, China Sep. 2021 – Jun. 2026

Ph.D. in Mathematics

Supervisor: Prof. Lun Zhang

Nankai University, Tianjin, China Sep. 2017 – Jul. 2021

B.Sc. in Mathematics and Applied Mathematics

RESEARCH INTERESTS

Random matrix theory; asymptotic analysis; Painlevé equations; universality

PUBLICATIONS

Asymptotics for the noncommutative Painlevé II equation

Junwen Liu, Luming Yao and Lun Zhang

Nonlinearity 39 (2026), 015017.

Large gap asymptotics of the hard edge tacnode process

Junwen Liu, Luming Yao and Lun Zhang

J. Lond. Math. Soc. 112 (2025), e70314.

HONORS AND AWARDS

Best Poster Award, The 2026 Joint Workshop of RMTA & SNAB 2026

National Scholarship (China) 2025

CONFERENCES

Invited Talks

Workshop on Integrable Systems and Random Matrix Theory Great Bay University, Jan. 4–16, 2026

Workshop on Integrable Systems and Random Matrix Theory Great Bay University, Jan. 5–17, 2025

Contributed Talks

Fifth ZiF Summer School: Randomness in Physics and Mathematics ZiF, Aug. 25–Sep. 6, 2025

AI for Math Graduate Academic Forum Shanghai Jiao Tong University, May 7–9, 2025

Other Conferences Attended

Universality, Nonlinearity, and Integrability in Honor of Percy Deift KIAS, May 12–16, 2025

Random Matrix Theory Summer School University of Michigan, Jun. 17–28, 2024

Random Matrices and Related Topics in Jeju Lotte Hotel, Jeju Island, May 6–10, 2024

TEACHING EXPERIENCE

School of Mathematical Sciences, Fudan University

Teaching Assistant, Basic Number Theory

Spring 2022

Teaching Assistant, Advanced Algebra I

Fall 2021

SKILLS

• **Computer Skills:** Mathematica, LaTeX, Microsoft Office

• **Languages:** Chinese (Native), English (Fluent)